Chapter 5.3 — Formula Sheet

Geometric Probability & 2D Integrals

Companion quick-reference to your Notebook 5 syllabus, your *20 Worked Examples* (Ex 1–20) and the *Stage-1 Hard Drill* (Q1–Q10)

A. Uniform on 2D Regions

Master Rule: Probability = Area Ratio [Ex 1; Q2]

$$(X,Y)\sim \text{Unif}(S): \quad f=\frac{1}{\text{Area}(S)}, \quad \mathbb{P}(A)=\frac{\text{Area}(A\cap S)}{\text{Area}(S)}$$

$$\mathbb{E}[g(X,Y)]=\frac{1}{\text{Area}(S)}\iint_S g\,dx\,dy$$

Draw S, shade A, *clip* any part of A that spills outside S. Compute the area first — it is the density.

Disk of Radius ρ: Radial Law [Ex 2; Q10]

$$\mathbb{P}(R\le r)=\frac{r^2}{\rho^2}, \quad f_R(r)=\frac{2r}{\rho^2}, \quad \mathbb{E}[R]=\frac{2\rho}{3}$$

Uniform in *area* is not uniform in *radius*. Polar: $dA=r\,dr\,d\theta$, e.g. $\iint_{\text{quarter disk}} x\,dA=\frac{1}{3}$.

Marginals = Slice Heights [Ex 3–4; Q2]

$$f_X(x)=\frac{\text{length of slice at } x}{\text{Area}(S)}$$

Disk: $f_X=\frac{2}{\pi}\sqrt{1-x^2}$. Triangle $0<y<x<1$: $f_X=2x$. Non-rectangular support $\Rightarrow$ *dependent*, even with flat density.

Shrinking Frame [Ex 5; Q1]

Unit square, $D=$ distance to nearest side:

$$\mathbb{P}(D>d)=(1-2d)^2, \quad \mathbb{E}[D]=\int_0^{1/2}(1-2d)^2 dd=\tfrac{1}{6}$$

Equidistant from a point and a line = parabola: curved boundary $\Rightarrow$ integrate (find crossings first).

B. Sums, Differences, Products, Ratios

$X+Y$ for iid $U(0,1)$: the Tent [Ex 6]

$$\mathbb{P}(S\le s)=\begin{cases} s^2/2, & 0\le s\le 1 \\ 1-(2-s)^2/2, & 1\le s\le 2\end{cases}$$

Density s then $2-s$. Corner triangles: legs s, or $2-s$.

$|X-Y|$ for iid $U(0,1)$ [Ex 7; Q4]

$$\mathbb{P}(|X-Y|\le d)=2d-d^2, \quad f=2(1-d)$$
$$\mathbb{E}=\tfrac{1}{3}, \quad \mathbb{E}[D^2]=\tfrac{1}{6}, \quad \text{Var}=\tfrac{1}{18}$$

Meeting Problem [Ex 8–9; Q4]

Window T, waits a,b: they meet iff the miss triangles are avoided:

$$\mathbb{P}(\text{miss})=\frac{(T-a)^2+(T-b)^2}{2T^2}$$

Equal waits: $1-(1-a/T)^2$. Rescale time to $[0,1]$ first; for $X\sim U(0,m), Y\sim U(0,n)$ clip lines against the $m\times n$ rectangle.

Products & Ratios [Ex 10; Q3, Q5]

$$\mathbb{P}(XY\le c)=c(1-\ln c), \quad 0<c\le 1$$

$$\mathbb{P}\left(\tfrac{X}{Y}\le t\right)=\begin{cases} t/2, & t\le 1 \\ 1-\frac{1}{2t}, & t>1\end{cases}$$

$$\mathbb{P}(k\le X/Y<k{+}1)=\tfrac{1}{2}\left(\tfrac{1}{k}-\tfrac{1}{k+1}\right)$$

Ratio = slope = wedge through the origin; tail $\frac{1}{2t}$ so $\mathbb{E}[X/Y]=\infty$. Line-vs-hyperbola: solve the crossing points, use them as limits.

n Uniforms: Simplex Volume [Ex 19]

$$\mathbb{P}(U_1+\cdots+U_n\le s)=\frac{s^n}{n!}\ (0\le s\le 1)$$

Symmetry $S\mapsto n-S$ gives the upper half. All $n!$ orderings are equally likely: $\mathbb{P}(X<Y<Z)=\frac{1}{6}$.

C. Stick-Breaking

Cut-Counting Law (n cuts, $n{+}1$ pieces) [Ex 12, 15; Q7]

$$\mathbb{P}(\text{piece} > t) = (1-t)^n, \quad \mathbb{P}(\text{all pieces} > t) = \big(1-(n{+}1)t\big)^n$$

(valid while the base is positive, else 0). Reserve t for every piece; the leftover *free* length is $1-(n{+}1)t$.

$$\mathbb{E}[\min] = \frac{1}{(n+1)^2}$$

Polygon / Triangle Test [Ex 11–12; Q7]

Polygon exists $\iff$ *every piece* $< \frac{1}{2}$ (no "bully"). Bully events are disjoint:

$$\mathbb{P}(\text{polygon}) = 1 - \frac{n+1}{2^n} \quad (n{=}2:\ \tfrac{1}{4},\ \ n{=}3:\ \tfrac{1}{2})$$

Ordered cuts $0 < a < b < 1$ live on a triangle of area $\frac{1}{2}$; valid region has area $\frac{1}{8}$.

Order Statistics of Pieces [Ex 13]

$$\mathbb{E}[k\text{-th largest}] = \frac{1}{n+1}\sum_{j=k}^{n+1}\frac{1}{j}$$

$n{=}2$: longest $\frac{11}{18}$, middle $\frac{5}{18}$, shortest $\frac{1}{9}$ (sum $= 1$).

Variants: How the Second Cut Is Chosen [Ex 14]

Break once, then break a second piece: *longer piece* $\to 2\ln 2 - 1 \approx 0.386$; *random piece (equal chance)* $\to \ln 2 - \frac{1}{2} \approx 0.193$; *piece chosen by length* (= two uniform cuts) $\to \frac{1}{4}$.

Inspection Paradox [Q6]

Piece containing a fixed point is size-biased:

midpoint: $\frac{7}{12}$, edge: $\frac{1}{3}$, random point: $\mathbb{E}\sum a_i^2 = \frac{2}{n+2} = \frac{1}{2}$

$\mathbb{P}(L > l) = 1 - (\frac{1}{2} + l)^2$ for the left end L of the midpoint's piece.

Acute Triangle [Ex 20]

Obtuse at c: $c^2 > a^2 + b^2 \iff b < \frac{1-2a}{2(1-a)}$, with $c < \frac{1}{2}$:

$$\mathbb{P}(\text{acute}) = 3\ln 2 - 2 \approx 0.0794$$

Given a triangle forms: acute ≈ 0.318, obtuse ≈ 0.682.

D. Capstones

Buffon's Needle [Ex 16; Q9]

Spacing d, length L; $x \le \frac{L}{2}\sin\theta$ on a $\frac{d}{2} \times \frac{\pi}{2}$ rectangle:

$$\mathbb{P}(\text{cross}) = \frac{2L}{\pi d} \ \ (L \le d), \qquad \mathbb{E}[\#\text{cross}] = \frac{2L}{\pi d} \ \text{ for all } L$$

$$L > d:\ \mathbb{P}(\ge 1) = \frac{2}{\pi}\left[\frac{L}{d}(1-\cos\theta_0) + \frac{\pi}{2} - \theta_0\right],\ \sin\theta_0 = \frac{d}{L}$$

Discriminant, Trading & Circle [Ex 17–18; Q8]

Real roots of $x^2 + Bx + C$: area under $C = B^2/4$ ($\frac{1}{12}$ for $U(0,1)^2$). Buyer/seller V_b, V_s iid $U(0,1)$: $\mathbb{P}(\text{trade}) = \frac{1}{2}$, $\mathbb{E}[G \mid \text{trade}] = \frac{1}{3}$. 3 points on a circle contain the centre: $\frac{1}{4}$ (cut the circle $\to$ stick). Chord: $\mathbb{E} = \frac{4}{\pi}$.

Monte Carlo Cost of an Area [Q10]

$$\hat{\pi} = 4\hat{p}, \quad \mathrm{Var}(\hat{\pi}) = \frac{\pi(4-\pi)}{n} \approx \frac{2.697}{n}, \quad \mathrm{SE} \le 10^{-3} \Rightarrow n \approx 2.7{\times}10^6$$

One-tool-tree reminder: uniform on a region means probability is area; every event is a shape inside a square, disk, triangle or simplex. Straight boundaries give triangles, curved boundaries (parabola, hyperbola) give one integral, and a stick with n cuts is just the simplex where "no piece $\ge \frac{1}{2}$" carves out the answer. Draw, shade, clip, measure.